Week 3 — Requirements Engineering & Mini-Capstone

> *"A PRD is a contract between the customer's problem and the engineering work that solves it. If either side cannot read it, the contract is void."*

Week 3 is the **first capstone week** of the academy. Quads convert their Week-2 strategy package — North Star, design principles, journey map, top feature concepts — into a single **Technical PRD** that an engineering lead would accept as input to scoping. The PRD ships Friday after a structured peer-review cycle.

This week is **explicitly non-AI**. The discipline of writing a PRD by hand, with your quad, is the muscle being built. AI returns in Week 4.

Learning outcomes

By Friday afternoon, each participant can:

1. Convert a customer problem into a structured **Technical PRD** with context, problem framing, goals, scope, AC, NFRs, risks, and open questions.
2. Write **granular, testable Acceptance Criteria** in Given/When/Then form that an engineer can implement against without a clarifying call.
3. Document **Non-Functional Requirements** across performance, security, accessibility, observability, and compliance — and defend each as a constraint, not a wishlist.
4. Conduct a structured **PRD peer review** that surfaces specific defects rather than general impressions.
5. Revise a PRD in response to peer review and produce a **review-resolution log** that shows which feedback was adopted, deferred, or rejected, with reasoning.

Daily map

Day	Topic	Key artifact produced
1	Drafting Technical PRDs	PRD Sections 1–3 (context, problem, goals, scope)
2	Writing granular Acceptance Criteria	8–12 testable ACs
3	Documenting Non-Functional Requirements	NFR section across 5 categories
4	Mini-capstone — PRD assembly (non-AI)	**Complete PRD** ready for review
5	PRD Review — morning reviews, afternoon revisions + secondary	**Revised PRD + review-resolution log**

The artifact is cumulative: each day's output is added to the PRD until Friday's revised version ships.

Daily cadence (applies Mon–Thu; Friday differs)

Clock	Block	Mix
09:00 – 09:15	Opening & objectives	Instructor
09:15 – 10:30	Teaching Block 1 + Activity 1	~25 min teach / 50 min quad work
10:30 – 10:45	**Break**	
10:45 – 12:00	Teaching Block 2 + Activity 2	~25 min teach / 50 min quad work
12:00 – 13:00	**Lunch**	
13:00 – 14:15	Teaching Block 3 + Activity 3	~25 min teach / 50 min quad work
14:15 – 14:30	**Break**	
14:30 – 16:00	Teaching Block 4 + Activity 4 + Wrap	~25 min teach / 45 min quad / 20 min wrap

Friday cadence (PRD review day)

Clock	Block	Mix
09:00 – 09:15	Opening; review pairings posted	Instructor
09:15 – 10:45	**Primary review round 1**	Two reviewing quads on each PRD
10:45 – 11:00	**Break**	
11:00 – 12:00	Author response + clarifying conversations	Author quads receive feedback
12:00 – 13:00	**Lunch**	
13:00 – 14:30	**Revisions**	Authors revise based on review
14:30 – 14:45	**Break**	
14:45 – 15:45	**Secondary review**	Different reviewers; read the revision
15:45 – 16:00	Wrap + sign-off	Final PRDs + resolution logs delivered

The "non-AI" rule (Mon–Thu)

This week is the discipline of **producing a defensible written artifact without a generative tool**. The reasoning:

- AI fills in vagueness with confident generic prose. Vagueness is the bug we're trying to eliminate.
- An engineer who reads an AI-drafted PRD often spots the generic prose and discounts the document.
- The TPM muscle being built is **structured thinking expressed as written words**. AI co-writing skips the thinking.

What's allowed: spell-check, dictionary, project notes, transcripts of customer interviews, your Week-1 and Week-2 artifacts. What's not allowed: generative AI for drafting, summarizing, or critiquing any part of the PRD.

The honor system holds. Friday's reviews surface generic prose quickly.

The PRD anatomy (the same template all

week)

```
# PRD — <Feature name>
**Author quad:** <names>  |  **Date:** <date>  |
**Status:** Draft / In review / Approved

## 1. Context
Why this work, why now. Customer signal. Strategy
fit.

## 2. Problem
The user's job-to-be-done; the friction they hit
today. Tied to your Week-1 problem statement and
Week-2 journey map.

## 3. Goals & non-goals
- Goals: what success looks like in user terms
- Non-goals: what we explicitly will not do (often
more valuable than the goals)

## 4. Scope (in / out)
The boundary. What ships, what doesn't, in this
iteration.

## 5. Solution sketch
Enough description that an engineer can imagine the
shape; not enough that you've designed it for them.

## 6. Acceptance Criteria
Given/When/Then form. Granular, testable,
falsifiable. (Day 2)

## 7. Non-Functional Requirements
Performance, security, accessibility, observability,
compliance. (Day 3)

## 8. Metrics & validation
Tied to the Tier Sheet from Week 2. How we'll know
it worked in 30 days.

## 9. Risks & open questions
Named honestly. The mark of a senior author.

## 10. Dependencies
Other teams, systems, decisions, data we need.

## 11. Out-of-scope follow-ups
Things this PRD acknowledges but doesn't ship.
```

Quads

Quads from Weeks 1–2 carry through. They authored together, they ship together. Friday's review pairings are different from authoring pairings (we cross-pollinate).

Friday review rubric

Reviewers score each PRD on:

Dimension	Weight	What "exemplary" looks like
Problem clarity	20%	Engineer could scope without clarifying call
AC testability	25%	Each AC is implementable + falsifiable; no "system should be intuitive"
NFR completeness	20%	5 categories present; each tied to a measurable target
Strategy linkage	15%	Goals tie to Week-2 NS / KPIs / journey friction
Risk honesty	10%	Real risks named; "no risks" is a fail
Writing discipline	10%	Clear, specific, no AI-generic prose

Bridge to Week 4

Week 4 turns from "what do we want" to "what's technically possible / constrained": monolith vs microservices, security & compliance, system mapping, latency/availability/rate-limit targets. Each quad's PRD becomes the input to the **technical architecture** conversation. The NFRs written this week are the **first draft** of the architectural constraints discussed Week 4.